BIO 005 Human Physiology · Week 7 notes · Dr. Sharilyn Rennie

The neuromuscular junction and contraction

The written notes for the the neuromuscular junction and contraction walkthrough. Week 7 competencies 1 to 5.

To print: these notes as a two-column PDF. Already printed the Week 7 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: from command to contraction

In your textbook: Silverthorn Chapter 12, the part on the somatic motor division and the neuromuscular junction, and Chapter 13, the skeletal muscle sections.

Dr. Rennie's videos for this part: Overview of muscle contraction.

At the neuromuscular junction, step by step

1. A somatic motor neuron, with its cell body in the spinal cord or brainstem, fires an action potential.
2. The action potential travels down the axon and its branches to an axon terminal on one muscle fiber.
3. It opens voltage-gated Ca2+ channels in the membrane of the axon terminal. Ca2+ flows in from the extracellular fluid.
4. The Ca2+ triggers docked synaptic vesicles to fuse with the membrane and release acetylcholine (ACh) into the synaptic cleft by exocytosis.
5. ACh diffuses across the cleft. Two ACh molecules bind to each nicotinic ACh receptor on the motor end plate, and its channel opens.
6. The channel is a nonspecific cation channel. Na+ flows in and a smaller amount of K+ flows out, so the end plate depolarizes: the end plate potential.
7. The end plate potential always reaches threshold (the safety factor). Voltage-gated Na+ channels next to the end plate open, and a muscle action potential spreads along the sarcolemma in both directions and down the T tubules.
8. Acetylcholinesterase in the cleft breaks ACh into acetate and choline. The receptor channels close, and the terminal takes the choline back in to make new ACh.

Excitation-contraction coupling, step by step

1. The muscle action potential travels down a T tubule, deep into the fiber.
2. It passes a DHP receptor, a voltage sensor in the T tubule membrane, which changes shape.
3. In skeletal muscle, the DHP receptor is physically linked to the ryanodine receptor (RyR1), a Ca2+ release channel in the membrane of the terminal cisterna. The shape change pulls it open.
4. Ca2+ flows out of the sarcoplasmic reticulum into the cytosol around the myofibrils. No Ca2+ from outside the cell is needed.

Calcium and the thin filament, step by step

1. Ca2+ binds to troponin C, part of the troponin complex on the thin filament.
2. Troponin changes shape and pulls tropomyosin aside.
3. The myosin binding sites on actin are uncovered.
4. Myosin heads bind firmly to actin and the crossbridge cycle runs for as long as Ca2+ stays bound.

One turn of the crossbridge cycle, step by step

1. Binding: the cocked myosin head, holding ADP and Pi, binds to actin.
2. Power stroke: release of Pi triggers the power stroke. The head swings and pulls the thin filament toward the M line.
3. At the end of the power stroke the head releases ADP and stays tightly bound to actin, with nothing on it. This is the rigor state.
4. Detachment: ATP binds to the head, and the head lets go of actin.
5. Reactivation: the head splits the ATP into ADP and Pi, and the energy re-cocks the head. It holds the ADP and Pi until it binds actin again.

Relaxation, step by step

1. The motor neuron stops firing, and acetylcholinesterase clears the ACh from the cleft.
2. No more muscle action potentials, so the ryanodine receptors close.
3. SERCA pumps in the SR membrane use ATP to move Ca2+ from the cytosol back into the SR.
4. Ca2+ comes off troponin, and tropomyosin slides back over the myosin binding sites.
5. The crossbridges stop. The sarcomeres return to resting length as the muscle is stretched by gravity or an opposing muscle and as titin springs back.

How a skeletal muscle is built

In your textbook: Silverthorn Chapter 13, the part on skeletal muscle anatomy.

Dr. Rennie's videos for this part: Skeletal muscle tissue, How connective tissue organizes a muscle, Overview of muscle contraction, Microscopic anatomy of skeletal muscle:

sarcolemma and sarcoplasm, Myofibrils, The structure of the sarcomere, A cross section of a sarcomere, Thick filaments and myosin, The sliding filament theory.

- Skeletal muscles are attached to bones by tendons and are controlled by somatic motor neurons.
- A whole muscle is wrapped in connective tissue and divided into fascicles, bundles of muscle fibers.
- A muscle fiber is one muscle cell: very long, with many nuclei, formed when many embryonic cells fused.
- Most of a fiber is filled with myofibrils, rods of contractile proteins that run the length of the cell.
- Each myofibril is a chain of sarcomeres, the contractile units, lined up end to end.

Sarcolemma

The cell membrane of a muscle fiber.

T tubules (transverse tubules)

Narrow tunnels of sarcolemma that dip deep into the fiber. They are open to the extracellular fluid and carry the action potential inside.

Sarcoplasmic reticulum (SR)

The muscle fiber's endoplasmic reticulum. A sleeve of tubes around each myofibril that stores Ca2+.

Terminal cisternae

The swollen ends of the SR, lying right beside a T tubule.

Triad

One T tubule with a terminal cisterna on each side.

Thick filament

Made of myosin. Its heads stick out toward the thin filaments around it.

Thin filament

Made mostly of actin, with tropomyosin and troponin lying along it. Anchored to a Z disc.

Titin

A huge elastic protein that runs from a Z disc to the M line. It holds the thick filaments in place and springs back after a stretch.

Z disc

The boundary at each end of a sarcomere. Thin filaments attach to it.

M line

The center of the sarcomere, where the thick filaments are held together.

Band	What is in it	When the sarcomere shortens
A band (dark)	The full length of the thick filaments, overlapping the thin filaments near its ends	Stays the same length
I band (light)	Thin filaments only, split by a Z disc	Gets shorter
H zone	The middle of the A band: thick filaments, no thin filaments	Gets shorter or disappears

- Sliding filament theory: neither filament gets shorter. Myosin heads pull the thin filaments toward the M line, so the filaments overlap more and the Z discs move closer together.
- The alternating bands are why skeletal muscle is called striated muscle.

At the neuromuscular junction

In your textbook: Silverthorn Chapter 12, the part on the somatic motor division and the neuromuscular junction.

Dr. Rennie's videos for this part: The neuromuscular junction, The muscle cell action potential, The action potential, part 1: ion channels and refractory periods, The action potential, part 2: the ion gates, Modulation of the neuromuscular junction.

Neuromuscular junction

The synapse between a somatic motor neuron and a skeletal muscle fiber: the axon terminal, the synaptic cleft and the motor end plate.

Motor end plate

The folded patch of sarcolemma under the axon terminal, packed with nicotinic ACh receptors.

Nicotinic ACh receptor

A channel that opens when two ACh molecules bind. It is a nonspecific cation channel: Na+ in exceeds K+ out, so the end plate depolarizes.

End plate potential

The depolarization of the motor end plate. It is a graded potential that happens only on the end plate.

Safety factor

In a healthy junction, the end plate potential is much larger than needed to reach threshold, so every motor neuron action potential produces one muscle action potential.

- Somatic motor neurons always release ACh and always excite the muscle fiber. No neuron inhibits a skeletal muscle fiber directly.
- Na+ has a much larger driving force than K+ at the resting potential, which is why more Na+ comes in than K+ goes out.
- Brain and spinal cord synapses usually need many inputs to add together to reach threshold. The neuromuscular junction is one for one.

What blocks it	Where it acts	How far the signal gets	The muscle
Botulinum toxin	Destroys the proteins vesicles need to fuse, inside the axon terminal	Action potential arrives and Ca2+ enters, but no ACh is released	Limp, flaccid paralysis
Curare	Binds the nicotinic receptors without opening them	ACh is released but cannot bind; no end plate potential	Limp, flaccid paralysis
Myasthenia gravis	Antibodies block and destroy nicotinic receptors	End plate potential is smaller and fails during repeated firing	Weakness that worsens with use and recovers with rest

What this means for patients

- Botulism from contaminated food causes dangerous paralysis. Tiny doses of botulinum toxin (Botox) treat some muscle spasms and chronic migraine and smooth wrinkles.
- Drugs related to curare relax the muscles during surgery while a machine breathes for the patient.
- Myasthenia gravis: during steady firing, a little less ACh is released with each action potential. With fewer receptors, later end plate potentials fall below threshold. Drooping eyelids and double vision are often the first signs.
- Myasthenia gravis is treated with acetylcholinesterase inhibitors, which let ACh stay in the cleft longer, and with treatments that calm the immune attack.

Excitation-contraction coupling

In your textbook: Silverthorn Chapter 13, the part on excitation-contraction coupling.

Dr. Rennie's videos for this part: Triads and terminal cisterns, Excitation-contraction coupling, Types of muscle tissue, Cardiac muscle tissue.

Excitation–contraction coupling
: The link between the muscle action potential (excitation) and the sliding of the filaments (contraction). The link is Ca2+ released from the SR.

DHP receptor
: A voltage sensor in the T tubule membrane, named for the dihydropyridine drugs that bind to it.

Ryanodine receptor
: A Ca2+ release channel in the SR membrane, named for ryanodine, a plant compound that binds to it.

	Skeletal muscle	Heart muscle
DHP receptor	A voltage sensor physically linked to the ryanodine receptor	A Ca2+ channel that lets Ca2+ in from outside the cell
What opens the ryanodine receptor	A mechanical pull from the DHP receptor	Ca2+ that came in through the DHP channel: calcium-induced calcium release
Ca2+ from outside the cell needed?	No	Yes
Without extracellular Ca2+	Still contracts	Does not contract

- T tubules carry the action potential deep into the wide fiber, right next to the terminal cisternae, so the whole fiber responds at once.

Calcium and the regulatory proteins

In your textbook: Silverthorn Chapter 13, the part on troponin, tropomyosin and the role of calcium in contraction.

Dr. Rennie's videos for this part: Thin filaments, troponin and tropomyosin.

- Each thin filament is two chains of actin twisted together. Every actin molecule has a myosin binding site.
- At rest, tropomyosin lies over the binding sites, and troponin holds it there. Myosin heads can only bind weakly.
- Ca2+ binds troponin C. Troponin changes shape and pulls tropomyosin off the binding sites. Myosin binds firmly and the power strokes begin.
- Ca2+ never binds actin or myosin here. Its target is troponin.

If this fails	Ca2+ in the cytosol	Crossbridges	The muscle feels
Ryanodine receptors will not open (nerve, end plate and ATP normal)	None released	Cannot bind; ATP keeps heads free	Limp
No ATP	Leaks out of the SR	Locked onto actin	Stiff (rigor)

The crossbridge cycle

In your textbook: Silverthorn Chapter 13, the part on the crossbridge cycle and the molecular basis of contraction.

Dr. Rennie's videos for this part: Crossbridge cycling and rigor mortis.

Stage	What happens	What is on the head after it
1. Binding	The cocked head binds to actin	ADP and Pi
2. Power stroke	Pi release triggers the power stroke; the thin filament moves toward the M line; then ADP is released	Nothing (rigor state)
3. Detachment	ATP binds, and the head lets go of actin	ATP
4. Reactivation	ATP is split into ADP and Pi, and the energy re-cocks the head	ADP and Pi

- ATP binds at detachment. ATP is hydrolyzed at reactivation. These are two different stages.
- The thin filament moves only during the power stroke.

- Each turn of the cycle uses one ATP per head. Heads along a thick filament cycle at different times, so the filament does not slip back.
- Without ATP, the heads finish the power stroke and stay locked onto actin: the rigor state. The muscle is stiff, not limp.

What this means for you

- Rigor mortis: after death, ATP runs out, Ca^{2+} leaks out of the SR, and the myosin heads lock onto actin. The body stays stiff until the muscle proteins begin to break down.
- Relaxation needs ATP to release each head from actin.

Relaxation

In your textbook: Silverthorn Chapter 13, the part on relaxation, and Chapter 12, the part on the neuromuscular junction.

Dr. Rennie's videos for this part: Muscle relaxation.

Shutoff job	Where	How	Uses ATP?	If it fails
Acetylcholinesterase (AChE)	Synaptic cleft	Splits ACh into acetate and choline; receptor channels close	No	ACh piles up: twitching, then weakness and paralysis
SERCA (sarco-endoplasmic reticulum Ca^{2+}-ATPase)	SR membrane	Pumps Ca^{2+} from the cytosol back into the SR, against its gradient	Yes	Ca^{2+} stays on troponin: the muscle stays contracted and cannot relax

- Clearing ACh stops new action potentials, but the muscle relaxes only when SERCA has removed the Ca^{2+}.
- Relaxation costs energy in two places: ATP to release each myosin head, and ATP for SERCA.

What this means for patients

- Organophosphates, in some insecticides and in nerve agents such as sarin, block acetylcholinesterase.
- At the neuromuscular junction, extra ACh first causes repeated firing and twitching. As the end plate stays depolarized, the nearby voltage-gated Na^{+} channels become inactivated, and the muscles become weak and paralyzed, including the breathing muscles.
- ACh also builds up at autonomic synapses: small pupils, drooling, sweating and a slow heart rate.
- Treatment includes atropine, which blocks the effects of extra ACh on the organs but does not help muscle weakness, and pralidoxime, which can restore acetylcholinesterase if given early.
- Brody disease, a rare inherited condition in which skeletal muscle SERCA works poorly, makes muscles stiffen and relax slowly after exercise.